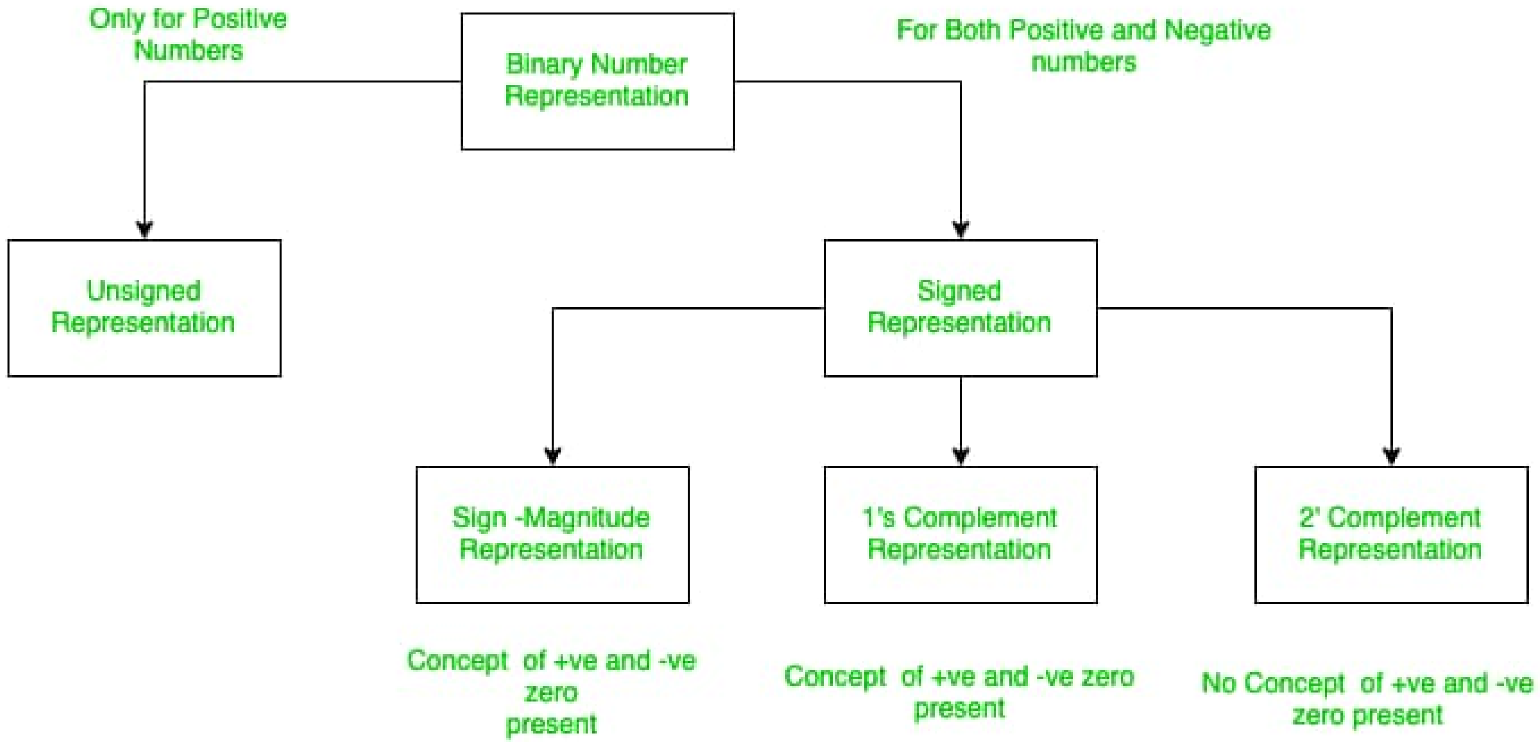
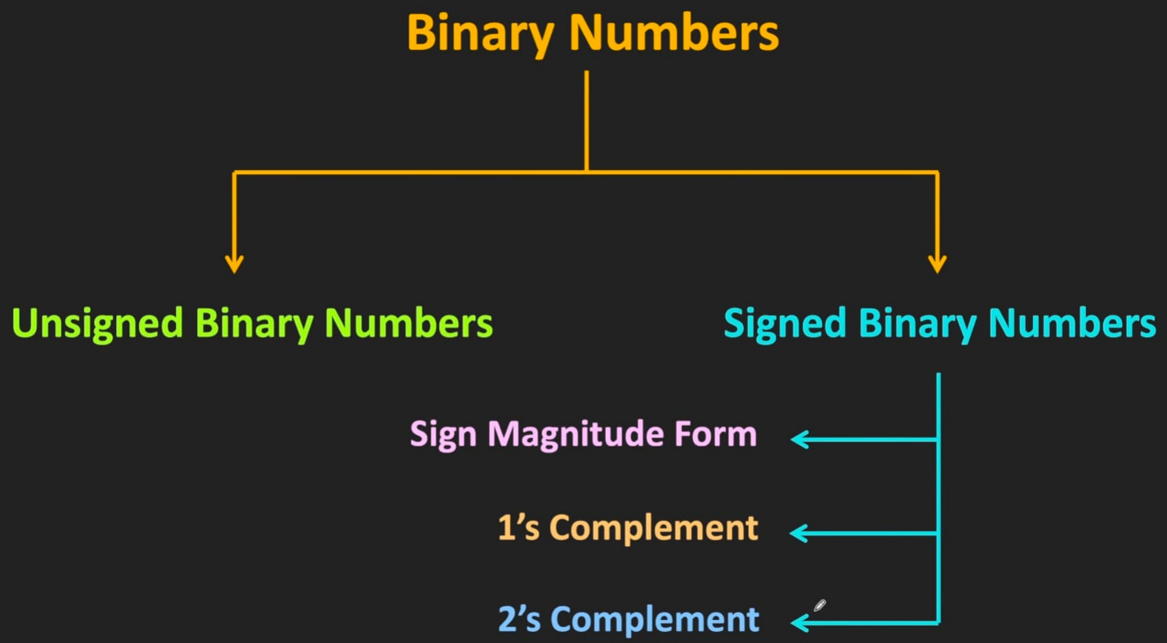

Only for Positive Numbers
Binary Number Representation
For Both Positive and Negative numbers
Unsigned Representation
Signed Representation
Sign -Magnitude Representation
1's Complement Representation
2' Complement Representation
Concept of +ve and -ve zero present
Concept of +ve and -ve zero present
No Concept of +ve and -ve zero present


Binary Numbers
Unsigned Binary Numbers
Signed Binary Numbers
Sign Magnitude Form
1’s Complement
2’s Complement

Signed Binary Numbers

- Represents both +ve & -ve numbers along with 0.
- MSB is a sign bit.
- For an N-bit singed number,

  MSB is a sign bit and remaining N-l bits are magnitude.

MSB

1 0 1 0 1 1 0 0

Magnitude

Signed Binary Numbers

Sign Magnitude Form

1's Complement

2's Complement

Representation of positive number is same in all three representation

Positive Numbers	MSB - 0
Negative Numbers	MSB - 1

Signed Binary Numbers

Sign Magnitude Form

Positive Numbers Sign Bit '0'

Negative Numbers Sign Bit '1'

+ 34 (in 8 bit)	0 0 1 0 0 0 1 0
- 34 (in 8 bit)	1 0 1 0 0 0 1 0

Signed Binary Numbers

Sign Magnitude Form

+ 52 (in 8 bit)	0 0 1 1 0 1 0 0
- 52 (in 8 bit)	1 0 1 1 0 1 0 0

Signed Binary Numbers

Sign Magnitude Form

+ 0 (in 8 bit) 0 0 0 0 0 0 0 0

- 0 (in 8 bit) 1 0 0 0 0 0 0 0

n- bit $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$

ALL ABOUT ELECTRONICS

Signed Magnitude Form

Decimal Numbers	Sign Magnitude Form
+7	0111
+ 6	0110
+ 5	0101
+ 4	0100
+ 3	0011
+ 2	0010
+ 1	0001
+ 0	0000
- 0	1000
- 1	1001
- 2	1010
- 3	1011
- 4	1100
- 5	1101
- 6	1110
- 7	1111

1's Complement Form

MSB

+ 34 (in 8 bit) 0 0 1 0 0 0 1 0

0 0 1 0 0 0 1 0

- 34 (in 8 bit) 1 1 0 1 1 1 0 1 1's complement of + 34

ALL ABOUT ELECTRONICS

1's Complement

n- bit $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$

4- bit $-(2^{4-1} - 1)$ to $(2^{4-1} - 1)$

-7 to $+7$

1's Complement Form

Decimal Numbers	1's Complement Form
+7	0111
+ 6	0110
+ 5	0101
+ 4	0100
+ 3	0011
+ 2	0010
+ 1	0001
+ 0	0000
- 0	1111
- 1	1110
- 2	1101
- 3	1100
- 4	1011
- 5	1010
- 6	1001
- 7	1000

2's Complement Form

MSB

+ 34 (in 8 bit) 0 0 1 0 0 0 1 0

0 0 1 0 0 0 1 0

- 34 (in 8 bit) 1 1 0 1 1 1 1 0 2's complement of + 34

ALL ABOUT ELECTRONICS

2's Complement Form

MSB

+ 34 (in 8 bit) $\quad$ 0 0 1 0 0 0 1 0

- 34 (in 8 bit)

$$
\begin{array}{r}
0\ 0\ 1\ 0\ 0\ 0\ 1\ 0 \\
1\ 1\ 0\ 1\ 1\ 1\ 0\ 1 \\
+ \qquad\qquad\qquad 1 \\
\hline
1\ 1\ 0\ 1\ 1\ 1\ 1\ 0
\end{array}
$$

1's complement of + 34

ALL ABOUT ELECTRONICS

2's Complement Form

- 60 (in 8 bit)

+ 60 (in 8 bit) 0 0 1 1 1 1 0 0

1 1 0 0 0 1 0 0 2's complement of + 60

2's Complement Form

n- bit -2^{n-1} to $(2^{n-1} - 1)$

4- bit -2^{4-1} to $(2^{4-1} - 1)$

-8 to $+7$

2's Complement Form

Decimal Numbers	2's Complement Form
+7	0111
+ 6	0110
+ 5	0101
+ 4	0100
+ 3	0011
+ 2	0010
+ 1	0001
+ 0	0000
- 0	-
- 1	1111
- 2	1110
- 3	1101
- 4	1100
- 5	1011
- 6	1010
- 7	1001
- 8	1000